

Concept Pitch

A classic remix of COD: Player clears sectors of zombies with limited ammo and select weapons where they will face occasional bosses and gain keys, used to unlock new sectors which will lead them to an intermission market. The player can spend money gained from kills collecting perks, upgrading weapons, and refilling ammo. Then they move onto the next sector, which are handcrafted and procedurally assembled, with more generated rooms and increasingly difficult zombies.

World Setting

Dystopian, abandoned world. Player setting is in an abandoned building, with long corridors and large rooms with obstacles that would appear in the aforementioned environment.

Artwork/Audio

Top down, old-school 2d click-and-shoot
Retro pixel textures

Audio is dystopian and fast-paced/excited

Core Gameplay

Game Systems (+ User Interface)

In Sector:

Player has health, stamina. Concepts:

- Health lost is permanent until player buys med kits at intermission, some health additionally restored by rare zombie drops

Player on default walks. When shift key is held, will sprint and use stamina. Concepts:

- Stamina replenishes automatically after sprint key is let go

Player has inventory of 3 guns. They each use their own ammo. They have magazines with select sizes. Guns can be sold or traded at intermission.

Zombies will spawn in calculated manner (see Progression Structure). There are different types of zombies, ex. Runner, Lobber, etc (we can be creative, ie zombies that blow up, zombies that are infected and deal poison)

In Intermission:

Player is presented with a shop. Concept:

- 1-3 guns
- 1-3 perks
- Upgrade gun slot
- Med kit slot

- Armor slot
- Loot box? (Random perk/gun)

Perks can go up in tiers. Each tier is exponentially more expensive (ideas: faster speed, ricochet bullets, faster reload)

Guns can be upgraded once for a fairly expensive price

Progression Structure

When entering new sections, zombie max hp and damage will increase. Every sector will have a max number of zombies spawned before granting player the key. When deciding what zombies to summon, game will decide what portions of total zombie pool will constitute each class:

Low-class zombie, medium-class zombie, high-class zombie, boss.

Technicals

Player Experience Goals

Basic controls are easy to understand, but difficulty increases fast. Addictive replayability, where player hops back in to hope to improve and learn more

Future Things to Look To

Multiplayer

Difficult to implement? Does it require a server? Would players need to make a login?

Milestones

Milestone 1

Player UI completely set up and player can shoot weapon with limited ammo. First Sector is naturally generated. Common zombies follow and try to kill player.

Milestone 2

Money collection system and shop is introduced after key is dropped from last zombie. Now, next sector can be created after shop is left and endless mode is created. Armor, Perks, Weapon upgrades added.

Milestone 3

More monsters and advanced zombie AI, bosses, progressive difficulty, refine all game aspects. Main menu screen with settings and Continue/New Run buttons. Sound effects

Milestone 4

Polish, Achievements, Statistics, Optimization, Accessibility, Controller Support, Steam integration

Additional Details

The game should prioritize:

- simple mechanics
- difficult gameplay
- meaningful progression
- high replayability
- satisfying combat

Player decisions should matter more than raw aim. The player should constantly feel under-equipped. Every shop visit should feel important.

Camera

Top-down. Orthographic camera. Camera follows player smoothly. Player always remains centered. No screen transitions. Entire room visible if reasonably sized. Controls

Movement

WASD.

Aim - Mouse Cursor

Shoot - Left Mouse

Reload - R

Sprint - Left Shift

Weapon Slot 1 - 1

Weapon Slot 2 - 2

Weapon Slot 3 - 3

Interact - E

Pause - Escape

Health

Maximum HP - 100. Health persists between sectors. Armor absorbs incoming damage before health. Armor only replenished through purchases.

Stamina

Used while sprinting. Maximum stamina: 100. Sprint drains stamina. Stops draining immediately when sprint released.

Recharge delay: 0.5 seconds

Recharge speed: 20 stamina/sec

Currency

Money dropped by zombies. Automatically collected within pickup radius. Money carries between sectors. Lost upon death.

Weapons

Slots may contain:

Pistol, SMG, Shotgun, Rifle, Sniper, Rocket Launcher, Special Weapons

Each weapon has:

Damage, Magazine Size, Reserve Ammo, Reload Speed, Fire Rate, Spread, Critical Chance, Piercing, Upgrade Level, Ammo Type, Every weapon uses separate ammo.

Reloads automatically if ammo is completely depleted and spare magazines are available

Weapon Quality

Common, Rare, Epic, Legendary

Higher rarity changes:

Damage, fire rate, reload, unique modifiers

Weapon Upgrades

Every weapon may be upgraded exactly once.

Upgrade increases:

Damage, Magazine, Reload Speed, Visual Effects. Cost scales with rarity.

Intermission Shop

Generated after every completed sector.

Contains:

1-3 random weapons, 1-3 random perks, Ammo refill. Armor, Med Kit, Weapon Upgrade, Mystery Box, Leave Sector

Player may remain indefinitely. No enemies.

Mystery Box

Random item.

Possible rewards:

Weapon, Ammo, Money, Armor, Healing

Small chance:

Legendary weapon, Perks

Perks

Remain until death

Examples:

Speed Boost, Reload Speed, Extra Health, Life Steal, Ricochet, Piercing Bullets, Fire Damage, Poison Bullets, Critical Chance, Double Money, Sprint Efficiency, Magazine Capacity, Explosion Resistance

Perk Tiers

Every perk has multiple levels. Level costs grow exponentially.

Zombie Classes

Common Zombie

Slow, Weak, Large numbers

Runner

Fast, Low HP, Medium damage

Tank

Slow, Huge HP, High damage

Lobber

Throws projectiles, Avoids player.

Exploder

Runs toward player. Explodes on death.

Poison Zombie

Leaves poison puddles.

Armored Zombie

Reduced bullet damage.

Necromancer

Revives nearby zombies.

Boss

Unique mechanics. One per boss encounter.

AI Behavior

Every zombie uses a finite state machine.

Idle, Roam, Investigate Noise, Chase Player, Attack, Dead

Gunshots attract zombies very well

Special zombies add unique states.

Sector Generation

Each sector generated procedurally.

Consists of:

Start Room, Combat Rooms, Hallways, Dead Ends, Treasure Rooms (rare), Boss Room (occasionally), Exit Door

Generation Rules

All rooms connected. No inaccessible spaces. Guaranteed path to exit. Random obstacle placement. Random decoration. No overlapping geometry.

Difficulty Scaling

Every completed sector increases:

Zombie HP, Zombie Damage, Spawn Count, Elite Spawn Chance, Boss Chance, Reward Multiplier

Spawn Director

Instead of spawning random enemies, use a budget system. Every sector has a difficulty budget.

Example

Budget = 100, Common Zombie = 1, Runner = 2, Tank = 8, Exploder = 5, Boss = 20

The director spends its budget until exhausted. This creates varied encounters.

Sector Completion

Sector ends when:

Spawn budget exhausted AND All zombies dead.

Reward:

Sector Key, Unlock Exit Door, Open Shop

Boss Encounters

Boss sectors every 5 sectors.

Bosses have:

Multiple attacks, Phases, Large health pools, Unique rewards,

Possible boss drops:

Legendary weapon, Rare perk, Large money bonus

Death

Upon death:

Run ends.

Display:

Sector reached, Money earned, Kills, Bosses defeated, Damage Taken, Favorite Weapon, Total Time

Player returns to Main Menu after being prompted with clicking button

No permanent progression in Version 1 (Rouge-like)

Save System

Mid-run saves. Save statistics, current sector, difficulty, player items and coins

User Interface

HUD, Health, Armor, Stamina, Money, Current Weapon, Ammo, Reload Indicator, Weapon Slots, Crosshair, Mini Objective, Sector Number, Pause Menu, Resume, Settings, Quit,

Audio

Weapon sounds, Reloads, Zombie screams, Boss themes, Footsteps, Ambient machinery, Impact sounds, UI clicks
Music intensity increases with combat.

Technical Architecture

Recommended Systems:

PlayerController, WeaponSystem, InventorySystem, HealthSystem, ArmorSystem, StaminaSystem, EnemyManager, SpawnDirector, RoomGenerator, Navigation, LootSystem, ShopManager, EconomyManager, PerkManager, AudioManager, SaveManager, GameStateManager, UIManager

Each system should be independent with minimal coupling.